Aviat NETWORKS

Eclipse

INTELLIGENT NODE UNIT

DATASHEET [ANSI]

FIPS VALIDATED 140-2 ™

[2]

The Eclipse™ Intelligent Node Unit (INU) is a highly modular and scalable indoor unit that delivers a unique combination of high capacity hybrid or all-packet transport, Carrier Ethernet/IP networking, and comprehensive Mission Critical Microwave features, enabling operators to prepare for the all-IP future.

Modular true hybrid system

A highly modular architecture which provides maximum protection of your investment. As your network and traffic requirements change over time, modules can be easily added to support additional radio directions or to change the traffic mix to facilitate the smooth migration from native TDM to advanced Carrier Ethernet/IP without any disruption.

Mission critical platform

Eclipse supports Mission Critical Microwave applications through a fully redundant architecture, with a variety of radio protection possibilities.

Strong security

Protection of your mission-critical data is imperative in today's world. Strong Security on the Eclipse platform is designed to provide peace of mind for those operators who need that extra level of security.

PROUDLY MANUFACTURED IN THE USA

FIPS 140-2 compliant, payload encryption, RADIUS, TLS1.2 support and software integrity check, enables up-to-date protection of your data and management traffic.

From native DS1, E1/T1 and OC3 to complete L2 Ethernet features with either SFP or electrical interfaces.

Redundancy and protection is assured on both Eclipse and radio levels with redundant cards, MHSB, FD and SD configurations.

System Parameters

General		Per link	Per node
Throughput/Capacity Range	Native Carrier Ethernet/IP [1]	10-940 Mbit/s	5.64 Gbit/s
	Native TDM	7 - 127x DS1	127x DS1
		1, 3, 4x DS3	6x DS3
		1, 2x OC-3	2x OC3
Fixed Modulation Options	Fixed		QPSK, 16, 32, 64, 256, 512, 1024, 2048 and 4096 QAM
Adaptive Modulation	Modulation Options		QPSK, 16, 32, 64, 256, 512, 1024, 2048 and 4096 QAM
Co-Channel Operation with XPIC	Optional		>20dB XPOL improvement
Adaptive Equalization			24 tap T/2 spaced feed-forward filter

Ethernet Specifications		
Ethernet Standards Compliance	Ethernet	IEEE 802.3
User Ports, per Data Access Card (DAC)		3x 10/100/1000BaseT, 1x 2500BaseT Electrical, 2x SFP Optical or Electrical
Networking Protocols		IPv4 and IPv6
Switch Capacity		5x 1Gbit/s user ports + 6x backplane ports
Maximum Frame Size		10000 bytes bi-directional
Throughput Acceleration	Frame Size Dependent	IFG & Preamble Suppression
Traffic Prioritization		Per port based prioritization
VLAN Support		IEEE 802.1Q, 802.1ad (Q-in-Q)
Flow Control		IEEE 802.3x
Link Aggregation		802.1AX LAG, L1LA (proprietary)
OAM		IEEE 802.1ag / ITU-T Y.1731
Monitoring	Port and Channel Status	Performance Graphs, RMON-1, Port and Channel

TDM Specifications		
Interfaces per Data Access Card (DAC)	NxDS1	1 to 16x 1.544 Mbit/s (DS1)
Multiple DACs of the same or different kind can be used per shelf	NxDS3	1 to 3x 44.736 Mbit/s (DS3)
	DS3 Mux	2x DS3 to 2x28x DS1 Mux, channelized
	NxOC3 Optical or Electrical	1 or 2x 155.52 Mbit/s (OC3)
	OC3 Mux	1x OC3 to 84x DS1 Mux

TDM Interface Standards	
DS1, DS3	ITU-T Rec. G.703, G.823
OC3, Electrical / Optical	ITU-T Rec. G.703, G.825 / ITU-T Rec. G.957, G.825

Protection	
Link Protection Options	Hot-Standby, Space or Frequency Diversity
Ring/Network Protection Options	Resiliant Wireless Packet Ring (RWPRTM), IEEE 802.1w RSTP ITU-T G.8032 ERP
User Line Interface Protection	1+1[3]: Ethernet, OC3, DS3, DS1

Security Compliance	
Security and Encryption	Optional FIPS 197 validated, 128/256-bit AES encryption Optional FIPS 140-2 Level 2 validated

Synchronization	
Synchronization Options	Synchronous Ethernet (G.8262)
	IEEE 1588v2 frames passed transparently
	DS1 Line clock

Standards Compliance	
EMC	FCC CFR 47 Part 15, ICES-003
Operation	EN 300 019, Class 3.2
Safety	UL 60950-1, UL 62368-1
NEBS	GR-1089-CORE, GR-63-CORE
Electric power substations	IEEE 1613

Mechanical, Environmental		
Plug-in card slots		INU: 4; INUe: 10
Rack Height		INU: 1RU; INUe: 2RU
Operating Temperature	Guaranteed	-10° to +55° C (14° to +131° F)
Humidity	Guaranteed	0 to 95%, non-condensing

Fault and Configuration Management		
Secure Management	Encryption	AES-128/192/256 3DES 168Bit
Local/remote Configuration Tool		Eclipse Portal
Network Management		Aviat Networks ProVision™

All specifications are typical values unless otherwise stated, and are subject to change without notice.

[1] Maximum Ethernet Throughput is for one 80MHz RF channel, single polarization, for 64 byte frame sizes. Corresponding Airlink base capacity is 366 Mbit/s.

[2] A Certification Mark of NIST, which does not imply product endorsement by NIST, the U.S. or Canadian Governments.

[3] Protection not supported in case of using INU (1RU) and RAC70.

Disclaimer

This material is for informational purposes only and does not constitute a legal obligation to deliver any product, feature or functionality and should not be relied upon in making purchasing decisions. All specifications are guaranteed values, at room temperature (20 to 30ºC, 68 to 86ºF), referenced to the ACU antenna port (including ACU losses) unless otherwise stated, and are subject to change without notice. The development, release and timing of any features or functionality described for our products is at Aviat Networks' sole discretion.

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